

Wiring

HC-05 Arduino

TXD D10

RXD D11 through divider

VCC 5V

GND GND

Divider construction

Arduino TX (5V)

|

1k Ω

|

+----- HC-05 RX

|

1k Ω

|

1k Ω

|

GND

After wiring, to reset Bluetooth, send this **Software serial code** below into your Arduino

Software serial code

```
#include <SoftwareSerial.h>
```

```
SoftwareSerial mySerial(10, 11); // RX, TX
```

```
void setup() {
```

```
  // Open serial communications and wait for port to open:
```

```
  Serial.begin(9600); //Baudrate for communication between computer and Arduino
```

```
  while (!Serial) {
```

```
    ; // wait for serial port to connect. Needed for native USB port only
```

```

}

//Serial.println("Goodnight moon!");

// set the data rate for the SoftwareSerial port
mySerial.begin(38400); //Baud rate for communication between HC05 module and Arduino board
//mySerial.println("Hello, world?");
}

void loop() { // run over and over
  if (mySerial.available()) {
    Serial.write(mySerial.read());
  }
  if (Serial.available()) {
    mySerial.write(Serial.read());
  }

  //Master 0024:07:004BED Baudrate:9600
  // Slave on hub address 5856:00:007A5F (which is 5856,00,007A5F)
  // Slave on truck

}
}

```

- Disconnect the HC05 RX and TX wires from the Arduino, and send your code to your Arduino
- Disconnect the Arduino for some seconds, (make sure serial monitor window is not opened on your computer)
- Hold the button on the HC05 while connecting back the Arduino to the computer. This is to put the HC05 into AT mode. Its blinking becomes slow.

- To be sure it has entered AT mode, type AT, Hold the button on HC05 and press send on the serial monitor window on your computer screen

NOTE: Some HC05 you might need to hold the button on them until you are done with all the settings, while some you just hold the button while plugging the Arduino USB into the computer. Just check your HC05 to know its style. After plugging your Arduino, it changes to blinking fast when you leave the button, then you need to do the process again, but this time you must hold the button until you are done with the settings.

Open serial monitor

➔ In your serial monitor, set baud rate to 9600, change “**No Line ending**” to both NL & CR

➔ Check baud rate the HC05 is currently on, type into the input box **AT+UART?** Hold the button on HC05 and press send on the serial monitor screen. You should see something like **OK**
+UART:9600,0,0

➔ To change baud rate, send **AT+UART=38400,0,0** Hold the button on HC05 and press send on the serial monitor window on your computer screen

➔ To check the role of the HC05, send **AT+ROLE?** Hold the button on HC05 and press send on the serial monitor window on your computer screen. You should get a result of **+ROLE:0 OK** or **ROLE:** something else. 0 is the role for a slave, and 1 is the role for a master.

➔ To change the role of the HC05 to master, send **= 1** Hold the button on HC05 and press send on the serial monitor screen. You should get OK

➔ To see and copy the address of the HC05 module so that you can link others to it, send **AT+ADDR?** Hold the button on HC05 and press send on the serial monitor window on your computer screen. You should get OK. You should get something like **+ADDR:0024:07:004BED OK**

➔ To bind master and slave, go to the AT mode of slave HC05 and copy the slave address, then go to the AT mode of master HC05, send **AT+BIND=0024,07,004BED** (assuming slave address is 0024,07,004BED). (i.e. in the ratio of 4:2:6)

To check whether a module is bound or where a module is bound to, type: **AT+BIND?**

➔ To unbind a Bluetooth module from any module it was previously bound with, send **AT+CMODE=1** This tells the module to connect to *any* available device, ignoring the bound address

➔ Copy that address and comment it into your code for reference so you don't forget, copy the baud rate too.

Configure Truck HC-05 #2 as MASTER

Disconnect SLAVE module.

Connect Truck HC-05 #2.

Enter AT mode again.

Set MASTER Mode

AT+ROLE=1

Expected:

OK

Set Fixed Connection Mode

AT+CMODE=0

Expected:

OK

Bind MASTER to SLAVE

Use the address you copied earlier.

Example:

If SLAVE address was:

98D3:31:F5A2C1

send:

AT+BIND=98D3,31,F5A2C1

NOTICE:

commas instead of colons

Expected:

OK

Set Same Baud Rate

AT+UART=9600,0,0

Expected:

OK

STEP 3 — Power Cycle Both Modules

Disconnect power from both.

Reconnect normally WITHOUT holding buttons.

STEP 4 — Automatic Pairing

After a few seconds:

- LEDs should slow blink together
OR
- become stable.

This means:

MASTER and SLAVE connected

STEP 5 — Your Arduino Code Already Works

This line in your truck code:

```
hubSerial.println(split2);
```

will now automatically transmit wirelessly to the hub HC-05.

And your hub code:

```
truckSerial.readStringUntil('\n');
```

will receive it.

Your architecture is now fully valid for the competition.